

	AFMS/AMMS	SLM-AFMS-AMMS
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SLM for Glider Life Extension Project

A/C : GENERIC

This document includes the Aircraft Flight Manual Supplement (AFMS) and the Aircraft Maintenance Manual Supplement for SLM operations.

It must be attached to the approved AIRCRAFT FLIGHT MANUAL (AFM) kept on board the aircraft.

The maintenance-related sections shall also be included in, or referenced by, the aircraft Aircraft Maintenance Manual.

LIST OF EDITION/REVISION INCORPORATED

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1. Scope

The device installed in the glider records in-flight acceleration data used to derive load-factor variations. These data are processed as part of the Structural Life Monitoring solution to calculate the measured usage spectrum and its Equivalent Fatigue Index (EFI). The EFI is compared with the approved reference EFI (EFI calculated using certification hypothesis). When the applicable EFI, inspection, and continued-airworthiness criteria are satisfied, the process may support operation beyond the originally certified safe-life limit.

2. Limitations

This supplement does not introduce new aircraft flight-envelope limitations.

Operation beyond the certified safe-life limit is permitted only when the approved SLM process and associated Instructions for Continued Airworthiness (ICA) are satisfied. Before the certified safe-life limit, flights without valid SLM data receive no measured-usage credit.

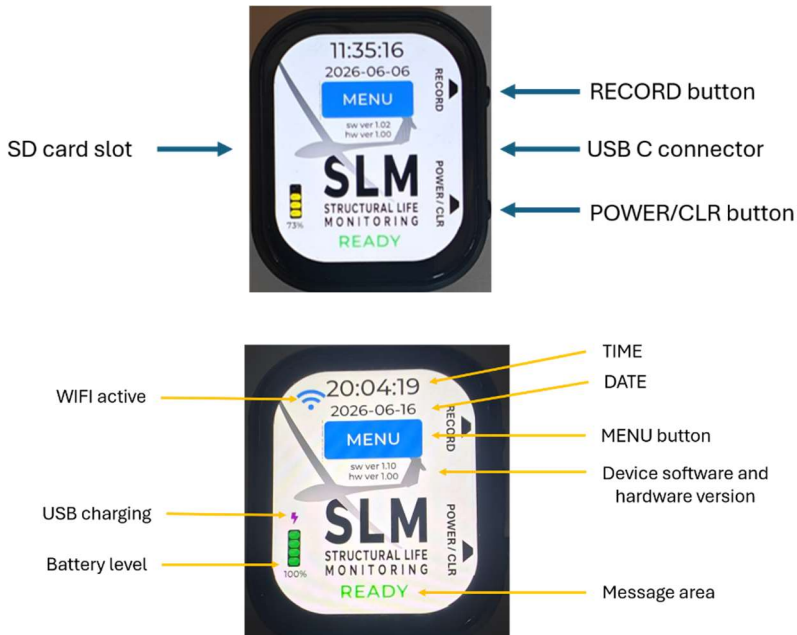
Beyond the certified safe-life limit, the SLM recorder, required calibration status, validated EFI status report, and required inspections are conditions for continued airworthiness.

3. Emergency procedures

There is no emergency procedure associated with this supplement.

4. Normal procedures

4.1. Recorder Description



The recorder is installed in the glider using a specific support (see installation instructions, section 7).

The USB-C connector should always be connected to the recorder.

All recorder controls are performed using the touch screen, the POWER/CLR and the RECORD buttons.

A 16GB Class 10 micro SD card is provided with the recorder.

4.2. Recorder Power Up

The recorder starts when the POWER/CLR button is pressed momentarily or when USB-C power is available.

There should be a green READY message at the bottom of the display.

/!\ If Amber or Red error messages are displayed at the bottom of the display, they should be corrected before flying (see message section 5).

/!\ If the recorder does not automatically start when the glider battery is connected, check USB-C and power connection.

4.3. Recorder Power Down

The recorder stops when the POWER/CLR button is pressed for ~3 seconds or when USB-C power is removed.

In some error conditions, this may require holding the button for up to 8 seconds.

When the recorder is in recording mode:

- if USB-C power is lost, the recorder continues to record.
- if the POWER/CLR button is pressed or the battery level is below 5%, recording will stop and recorder will power down.

4.4. Recorder Special Screens

When there is no interaction with the recorder for more than 10 seconds, the display goes to stand-by mode to save battery power. Display can be activated by touching the screen.



When battery level goes below 5%, the recorder display a red message asking the user to connect active USB-C power, before shutting down.

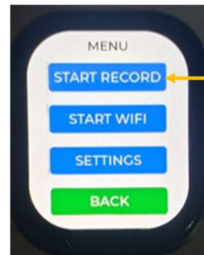


4.5. Start Recording Mode

The recorder can be started by pressing ~1 second on the hardware RECORD button or on the display MENU -> START RECORD button.



RECORD
button



START
RECORD
button

The recorder should display STARTING and immediately after, RECORDING



The recorder creates one daily binary file. Additional same-day recording sessions are appended to the same daily file, and the suffix is updated to show the number of recording sessions contained in the file, for example FCJAF_20260616_1.bin, then FCJAF_20260616_2.bin.

4.6. Stop Recording Mode

The recording mode is stopped by:

- pressing on the hardware RECORD button for ~3 seconds



- pressing on the display red STOP RECORD button for ~1 second (MENU page)



- pressing the POWER/CLR button for ~3 seconds *
 - if the battery level goes below 5% *
- * : in this case the recorder will also power down.

4.7. Pre-Flight Check

When the glider's battery is installed and connected, the recorder should automatically start and the display should be active with a green READY message at the bottom of the display.

/!\ If the recorder does not automatically start when glider battery is connected, check USB-C and power connection.

/!\ If Amber or Red error messages are displayed at the bottom of the display, they should be corrected before flying (see message section 5).

4.8. Before Flight

Start recording mode.

/!\ If a flight is performed without SLM recording, or if the recorded data is invalid, the flight shall be treated in accordance with the approved missing-data procedure and No measured-usage credit is granted for that flight.

4.9. After flight

Stop the recording mode.

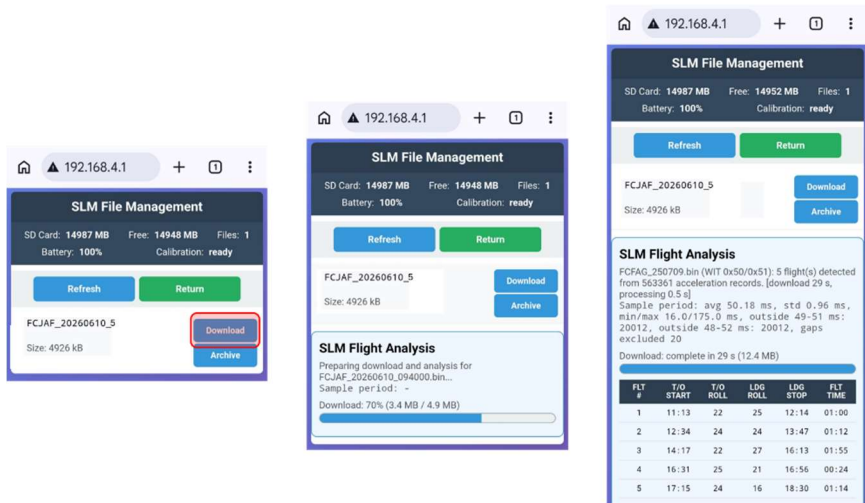
4.10. At the end of flying day

The last pilot-in-command to fly the glider must update the glider's usage information on the Structural Life Monitoring (SLM) website.

Using a device capable of WIFI (a smart phone or a PC), the files should be first downloaded from the recorder to the device and then transferred to the SLM website.

See section 6.3 for access to the recorder Web pages.

Select File Management.



Selecting Download will transfer and process the selected file from the recorder to the device.

When a file is transferred to the device, it is necessary to upload it to the main SLM server using the address:
<https://planeur-la-llagonne.com/gestion-vols/index.php>

You need an account and a password to access the SLM server (contact support to get an account)

Prolongation de la durée de vie des planeurs

Connexion

Identifiant

Mot de passe

Se connecter

Comptes disponibles :

- ✓ Planeur F-CFAG
- ✓ Planeur F-CJAF
- ✓ Administrateur

When the account and password are entered, you have access to the main SLM server page

Prolongation de la durée de vie des planeurs Planeur : F-CFAG Déconnexion

Enregistrer un nouveau vol

Date du vol *

Nom du pilote *
 Ex: Jean Dupont

Fichier de vol (.bin) *
 No file selected.

Enregistrer le vol

Mes vols enregistrés (0)

Aucun vol enregistré pour ce planeur.

To upload a file to the SLM server, you need to enter the date, the pilot name and select a file previously downloaded from the recorder to the device.

When the file is correctly uploaded to the SLM webserver, it can be archived from the recorder by selecting Archive on the SLM Recorder web page.

/!\ The recorder only accepts 50 files.

When all operations associated with the recorder are finished, the recorder should be stopped.

5. Messages

Message	Type	Operator action
BOOT	Startup transient	
READY	normal operation	
RECORDING	normal operation	
STARTING	recording start transient	
STOPPING	recording stop transient	
SHUTDOWN	Shutdown transient	
SD OK/CLR	SD recovered / clear prompt	press clear to confirm SD error correction
NEED SETTINGS	setup required	perform date, time, registration and password SETTINGS
ACC CAL REQ	accelerometer calibration required	perform Web acceleration calibration menu (START WIFI)
INST CAL REQ	installation calibration required	Perform Web installation calibration menu (START WIFI)
NO SD	SD not detected warning	insert a valid SD card
SD LOW	SD memory low	insert an SD card with enough free space (500 MB min.)
SD FULL (FILES)	SD has too many files (50 max.)	download/delete files from recorder Web interface (START WIFI)
LOW BATT	Low power warning	connect USB
CAL FAULT	calibration fault	perform factory reset and perform new calibrations
FATAL WDG/CLR	watchdog fault acknowledgement	press the power/clear button to acknowledge/clear error *

ACCEL ERR	hardware/runtime error	press the power/clear button to stop recorder, then restart *
RTC ERROR	hardware/runtime error	press the power/clear button to stop recorder, then restart *
PMU ERROR	hardware/runtime error	press the power/clear button to stop recorder, then restart *
REC FAIL	recording failure	press the power/clear button to stop recorder, then restart *
TOUCH ERROR	hardware/runtime error	press the power/clear button to stop recorder, then restart *
ERROR	generic fallback error	press the power/clear button to stop recorder, then restart *
SD ERROR	SD/storage fault	press the power/clear button to stop recorder, then restart *
GENERIC ERROR	fatal error	press the power/clear button to stop recorder, then restart *

* It might require pressing the power/clear button up to 8 seconds to stop device.
 * If the error persists after restart, report problem to support

6 Recorder Special Functions

6.1 Recorder Reset

!\\ when resetting the device, all SETTINGS and CALIBRATIONS parameters are lost.

To reset device, press the RECORD button and immediately press the POWER/CLR button until the device stops.

When the device is restarted, it should ask the user to configure SETTINGS and CALIBRATIONS.

Corresponding amber messages are displayed.

To help the user navigate through the configuration pages, the required button are in Amber color.

Buttons return to their normal color when all required configuration changes are done.

6.2 Recorder settings

!\\ the recorder cannot be operated until all settings have

been performed.

From the main page with the device ready, select MENU then SETTINGS.

The SETTINGS page allows user to set DATE, TIME, REGISTRATION and WIFI PASSWORD.



The values are adjusted using the rollers.

When values are set correctly, select SAVE.

If you are on a setting page by mistake, select BACK.

When settings need to be updated, amber messages should be displayed on the main screen, to indicate which setting should be updated.

The device also turns amber the buttons which lead to the required setting. When all required settings have been configured, messages and buttons are back to their normal color.

6.3 Access to the recorder Web pages

Using a device capable of WIFI connection (a smart phone or a PC), the recorder file-management, maintenance, calibration, and firmware-update pages are accessed from the recorder local web page.

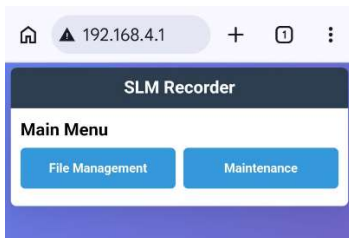
To connect to the recorder WIFI and Website, select MENU and START WIFI.



START
WIFI
button

Open the WIFI connection manager on your device and select the recorder WIFI SSID which is SLM- followed by the glider registration.

Then open a browser in incognito mode on the device and type <http://192.168.4.1/>



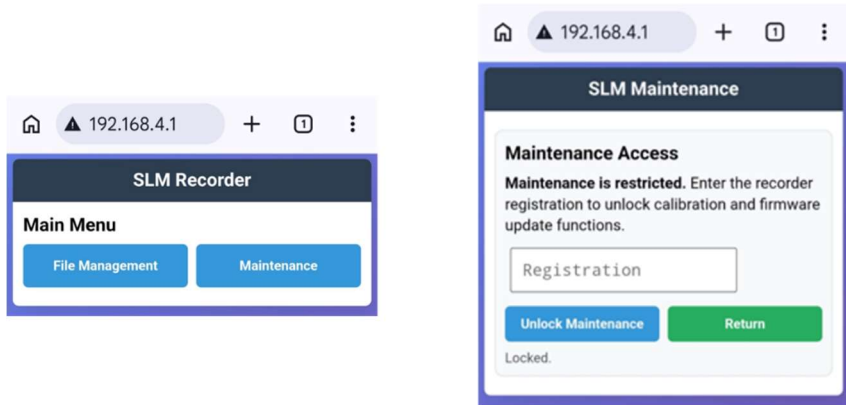
When the WIFI activities are finished, you need to stop the WIFI by selecting the STOP WIFI in the MENU.

6.4 Initial calibrations

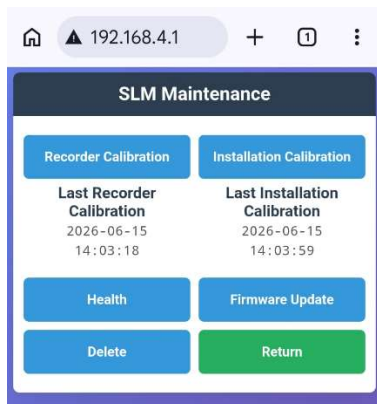
Access to the Maintenance/Calibration pages requires a password.

Access the recorder Web page (see section 6.3)

Select the Maintenance menus.



Enter the password to unlock maintenance, then select the required calibration



/!\ when previous calibrations have been performed, the calibration date is indicated below the button.

/!\ When calibrations need to be updated, amber messages should be displayed, to indicate which calibration should be updated.

/!\ The device also turns amber the buttons which lead to the required calibration. When all required changes have been performed, messages and buttons are back to their normal color.

6.4.1 Recorder calibration

This is an internal recorder calibration to ensure the acceleration sensor performance is appropriate.

Select the Recorder Calibration.

SLM Recorder Calibration

Workflow: Start calibration, place the recorder still on each of its six faces, and wait for each face to show OK. The recorder automatically keeps the best capture for each face. It is good practice to leave the recorder on a given face until the last best update is more than 10 seconds old. Save calibration when all six faces values are satisfactory.

Start

Save

Cancel

Return

Status: valid since 2026-06-15 14:03:18
Samples processed: 0
Lowest stddev: -
Last best update: -
Faces:

+X	-	-X	-
+Y	-	-Y	-
+Z	-	-Z	-

Axis	Gain	NVS Gain	Offset	NVS Offset
X	-	0.999088	-	7.0
Y	-	1.001089	-	-43.9
Z	-	1.000876	-	-16.2

Install the recorder in its calibration support (reference SLM-CAL01).



Start the calibration process and follow the workflow.

During calibration, each face is initially not validated, turns amber while the corresponding face is being acquired, and turns green/OK when a satisfactory stable capture has been obtained.

When all six faces have been calibrated, select SAVE.
If there is a problem, select CANCEL and start the process again.

The date of the recorder calibration should be entered in the aircraft logbook

6.4.2 Installation calibration

This calibration compensates the installed orientation of the recorder so that the corrected vertical acceleration corresponds to the aircraft load-factor axis when the glider is

in the defined level-flight attitude.

Select the Installation Calibration menu.

SLM Installation Calibration

Workflow: Put the glider in its flight-level attitude with wings leveled, following the AMM procedure. Sensor calibration must already be valid before attempting installation calibration. Click Start, leave the glider still. It is good practice to wait until the last best update is more than 10 seconds old. Save calibration when noise is satisfactory.

Start

Save

Cancel

Return

Status: valid since 2026-06-15
14:03:59
Samples processed: 0
Lowest noise: -
Last best update: -

```
[ -0.025291 -0.013315 -0.999591 ]
[ -0.013315 0.999827 -0.012981 ]
[ 0.999591 0.012981 -0.025463 ]
```

The glider should be positioned in the flight level attitude with wing leveled (same Aircraft Maintenance Manual procedure required when performing Weight & Balance).

Start the calibration process and follow the workflow.

When calibration is correct, select SAVE.

If there is a problem, select CANCEL and start the process again.

The date of the installation calibration should be entered in the aircraft logbook.

6.5 Recorder Health

By selecting Maintenance / Health, the user can access the recorder diagnostics page.

6.6 Firmware Update

/!\ Firmware update should only be attempted if instructed by SLM support.

Access the recorder Web page.

Select Maintenance / Firmware Update



The new firmware file should be provided by SLM support (with a link to download the file) and uploaded to the device which is used to connect to the recorder.

Select the Browse / file select button and select the firmware file which is on the device.

Select Upload Firmware and wait for device to be updated.

Check the device display to verify that the device has

restarted.

/!\ After a firmware update, the device restarts in the normal READY condition and WIFI is OFF. You need to activate WIFI on recorder and connect your phone or PC if you want to access WEB services again.

/!\ Following a firmware update, it is possible that settings and calibrations have to be performed. This can happen when the firmware changes the way settings and calibrations are stored in memory.

/!\ When settings and calibrations need to be updated, amber messages should be displayed, to indicate which parameter should be updated.

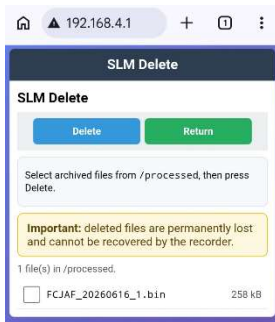
/!\ The device also turns amber the buttons which lead to the required changes. When all changes have been performed, messages and buttons are back to their normal color.

The date and reference of the firmware should be entered into the aircraft logbook.

6.7 File Delete

When files are archived using SLM File Management, they are moved to the “/processed” folder of the SD card. If necessary, the files can be deleted from the SD card.

Selecting Maintain / Delete provides access to the Delete function.



User should select which file to delete before pressing Delete.
 /\ This operation cannot be undone!

7 Recorder Installation

/\ When all the installation steps have been performed, the installation details should be entered in the aircraft logbook using appropriate procedure.

/\ Before starting the installation, it is recommended to reset the settings (see section 6.1) and configure the recorder with the appropriate date, time, aircraft registration and WIFI password in the SETTINGS menu.

7.1 Installation of the recorder in the glider

The recorder is provided with an installation support (reference: SLM-SUP01).

The support should be installed in the fuselage, in a rigid area close to the Center of Gravity and wing spar ($< \sim 0.3\text{m}$), with suitable access for operation of the recorder and routing of the USB-C cable (both the buttons and the USB-C socket are

located on the right side of the recorder).

/!\ the recorder should not be in an area where there is a risk of object touching it.

/!\ the recorder should be installed with strong dual sided mounting tape. The weight of the recorder and its support is ~40g, the attachment of the support should be tested to verify if can withstand 640g of forward longitudinal force.

7.2 Electrical wiring and power supply installation

The recorder is provided with a small power supply (reference SLM-POW01).

The power supply electrical wires should be connected to the glider's battery connector and protected with a 1A fuse. The power supply USB-C line should be connected to the recorder USB-C socket.

/!\ Everything should be secured to ensure there is no electrical risk. The USB-C line should also be secured close to the recorder to make sure it cannot introduce vibration on the recorder.

7.3 Initial recorder calibration

Before installing the recorder in the glider on its support, it should be calibrated (see section 6.4.1).

7.4 Installation calibration

When the recorder is installed in the glider, it is necessary to perform an installation calibration (see section 6.4.2).

8 Maintenance

During the annual visit, a recorder calibration must be carried out and the status report of the Equivalent Fatigue Index (EFI) usage statistics should be filed.

The calibration results are a condition for validating the recordings uploaded since the previous calibration.

The EFI status report provides the information required to allow aircraft return to service.

The date of the calibration and the EFI status report shall be included in the aircraft maintenance documentation.

The glider Aircraft Maintenance Program should be updated to reflect these requirements.

8.1 Recorder Calibration

The calibration shall be performed at least once a year but can be repeated as many times as desired.

The process to calibrate is described in section 6.4.1

8.2 Equivalent Fatigue Index documents

/!\ The recorder calibration should be performed before generating the EFI status report. Flights performed after the last valid calibration and before the EFI report may be treated as non-validated or handled in accordance with the approved data-processing procedure.

The EFI report provides the information required for the Go/No-Go decision supporting the release to service.

9 Performance

There are no performance changes associated with this supplement

10 Weight and balance

Due to very light weight and location (less than 100g near C.G. for the complete system), is it not necessary to perform a physical W&B update. However, the equipment should be added to the glider equipment list.

11 Hardware delivered with recorder

- 1x Recorder SLM-REC01
- 1x Installation support SLM-SUP01
- 1x Calibration support SLM-CAL01
- 1x 12V to USB-C power supply SLM-POW01

12 In case of problem

Contact support :
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